COMPUTER VISION

Light, colour and images

Session 02

From camera measurements
to histograms and image enhancement

Mohammad Sabokrou

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Light and the object surface

Illumination = lighting

The lamp shines light on the mug. The lamp is the light source.

Surface = the outside

The mug’s ceramic outside is its surface. Light hits this surface.

Reflection = returning light

Some light leaves the mug’s surface and reaches the camera.



A warm lamp can make a white mug look yellow.

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Optics: how a lens forms an image

Optics = how light behaves

In a camera, optics explains how the lens bends light to form a picture.

Lens = the curved glass at the front

It directs light onto the sensor inside the camera. The sensor sits at the image plane.

Focus = placing the image plane correctly

The lens brings rays from one scene point together on the sensor. In focus: they meet in a small spot, so edges look sharp. Out of focus: they form a larger circle, so nearby pixels mix and look blurry.





In focus: sharp mug



Out of focus: blurry mug

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The sensor: from light to numbers

Sensor = a light-sensitive chip

It sits inside the camera, behind the lens. It contains a grid of tiny light-measuring sites.

Each site collects light

During the exposure, captured light produces electrical charge. The camera measures this charge.

Electronics turn readings into numbers

The image stores pixel values in rows and columns. A pixel is one small part of the picture.

8-bit grayscale: 0 = black, 255 = white.



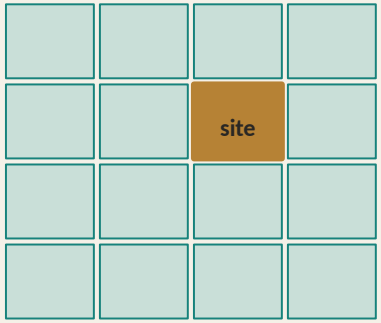
The coloured rectangle inside is the sensor.

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How tiny sensor sites create an image

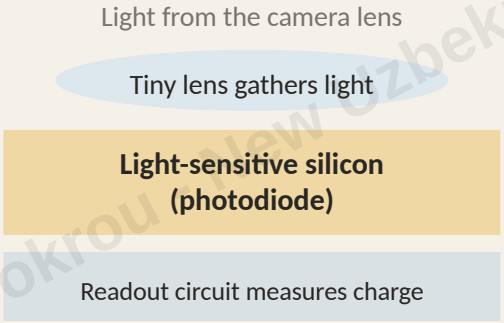
Simplified grayscale sensor: one site measures one small area of the image.

1. A grid of sites



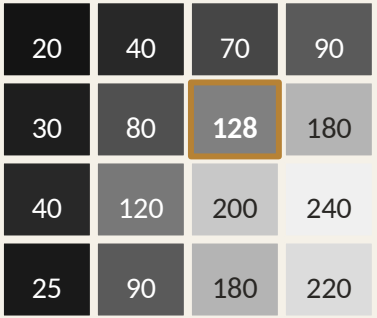
The lens forms a picture on the sensor. Each square is a tiny measuring site, also called a photosite.

2. Inside one site



Absorbed light creates electrical charge. Charge builds up during the exposure, the time spent collecting light.

3. A grid of numbers



The charge becomes a voltage. An analogue-to-digital converter (ADC) compares it with fixed voltage levels and assigns an integer, such as 128. The grid of integers is the image.

Example values, not measurements. In many colour cameras, red, green or blue filters cover sites; software combines nearby readings into colour pixels.

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How light becomes a measured number

The sensor measures an electrical response to light. Here is where the number comes from.

1. Light arrives

Light comes in tiny packets called photons.
They reach a sensor site while the camera collects light.

2. Charge builds up

Absorbed photons can release electrons in the silicon.
The site collects these electrons as electrical charge.

3. Measure a voltage

Electronics convert the collected charge into a voltage.
Suppose the measured signal is 0.20 volts (V).

0.20 V

4. Convert to a number

A digital converter assigns a number to the voltage.
In our example, 0–1 V maps to the numbers 0–255.

51

Example calculation: 0.20 V ÷ 1.00 V × 255 = 51

51 is a digital code, not a light-intensity unit. This simplified 8-bit example uses an illustrative voltage range; real cameras may process the code further.

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OPENING QUESTION · 2–5 MIN

When did the grey card become blue?

A lamp shines on the grey card. The card reflects some light toward the camera. Different lamps can make the same grey card look warmer or cooler.

PREDICT

Did the card change, or did the light shining on it change?

A photo depends on both the object and its lighting.



Four questions for today

1. LIGHT AND NUMBERS

How does light reaching a camera become the numbers in an image?

2. COLOUR

Why can the same white mug look yellow under a different lamp?

3. BRIGHTNESS

How can a graph count the dark and bright pixels? This graph is called a histogram.

4. SIMPLE IMAGE CHANGES

What happens when we make a photo brighter? Which details become easier or harder to see?

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Same cup, different pixels: what changed?

“Object change” means the cup itself changed: its shape, material, colour, position or damage. Here the cup stays the same; only the lighting changes.



Same cup under warm light



Same cup under cool light

Pixel values can change because of the object, lighting, viewpoint or camera settings.

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From a physical scene to a distribution of values

0–20

FORMATION

Light, surfaces, optics, and sensor

20–50

COLOUR

RGB, intensity, and colour spaces

50–100

HISTOGRAMS

Exposure, contrast, and equalisation

100–120

REVIEW

Explain image changes and compare examples

Prediction, measurement, explanation, and repair

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The same object can create different image arrays



The cup keeps its identity. The recorded array changes when illumination, viewpoint, scale, occlusion, or sensor noise changes.

How a surface changes light



The lamp supplies light. The paper reflects some of it; some reflected light reaches the camera.

$$L_r(\lambda) = E(\lambda)\,\rho(\lambda)$$

Reflected light = incoming light × reflected fraction

λ (lambda): wavelength

Which light we mean, such as red light at 650 nm.

E(λ): incoming light

How much light at that wavelength reaches the paper.

ρ(λ) (rho): reflectance

The fraction reflected: 0.8 means 80%. It ranges from 0 to 1.

$L_r(\lambda)$: reflected light

The result after reflection. The small r means reflected.

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At one red wavelength: 100 relative units × 0.8 = 80 reflected units.

Illustrative values. Simplified model for a matte surface, with fixed geometry; these are not pixel values.

Where does a point appear in the image?

$$u = \frac{fX}{Z} \qquad v = \frac{fY}{Z}$$

What the symbols mean

f: focal length, the lens's distance scale.
X, Y: the point's horizontal and vertical offsets from the camera's central viewing line.
Z: how far forward the point is.
u, v: its position in the image, measured from the image centre in this simple model.

Example: all distances are in millimetres

f = 50 X = 100 Y = 50 Z = 1000
The point is 1 metre in front of the camera.

$$u = \frac{50 \times 100}{1000} = 5 \text{ mm}$$

$$v = \frac{50 \times 50}{1000} = 2.5 \text{ mm}$$

The point appears 5 mm across and 2.5 mm up from the image centre. These are distances on the image plane, not pixel indices.

Twice as far away: Z = 2000 mm gives u = 2.5 mm and v = 1.25 mm.

With X, Y and f unchanged, doubling depth halves the image offsets. Ideal pinhole model; lens distortion is ignored.

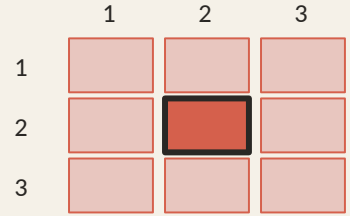
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The meaning of each term in the sensor formula

$$I_c[x, y] = Q(g\, s_c(x, y) + n_c)$$

One image location

x = column; y = row



At (2, 2), c selects a channel.
c is a label, not a multiplier.

R = 105 **G = 60** **B = 30**

RGB = (105, 60, 30)

s: sensor signal

Electrical measurement of the light at this location and colour channel.

g: gain

A multiplier that amplifies the signal. For example, g = 2 doubles it.

n: noise

Unwanted variation in the reading. It can increase or decrease the result.

Q: digital conversion

Maps the reading to an allowed integer. I is that stored value at [x, y], channel c.

Put the red numbers into the equation:
gs + n = 2 × 0.20 + 0.01 = 0.41 V
Q(0.41) = round(0.41 ÷ 1.00 × 255) = 105

So I_R[2,2] = 105. Green and blue use their own sensor readings. Values are simplified examples.

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A camera records light after the world transforms it



$$I(x, y, c) = Q[gE(x, y, c) + n]$$

Spectral power = amount of each light colour. E = light reaching the sensor; g = electronic multiplier (g = 2 doubles the signal); n = unwanted error; Q = conversion to a stored integer.

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Exposure: how much light the sensor collects

Exposure = collected light

How much light reaches the sensor during a photograph.

Underexposed = too little light

The photo is too dark for the detail we want. Details in dark areas become hard to see.

Overexposed = too much light

The photo is too bright. Bright areas can become plain white and lose detail.

Same scene, different exposure



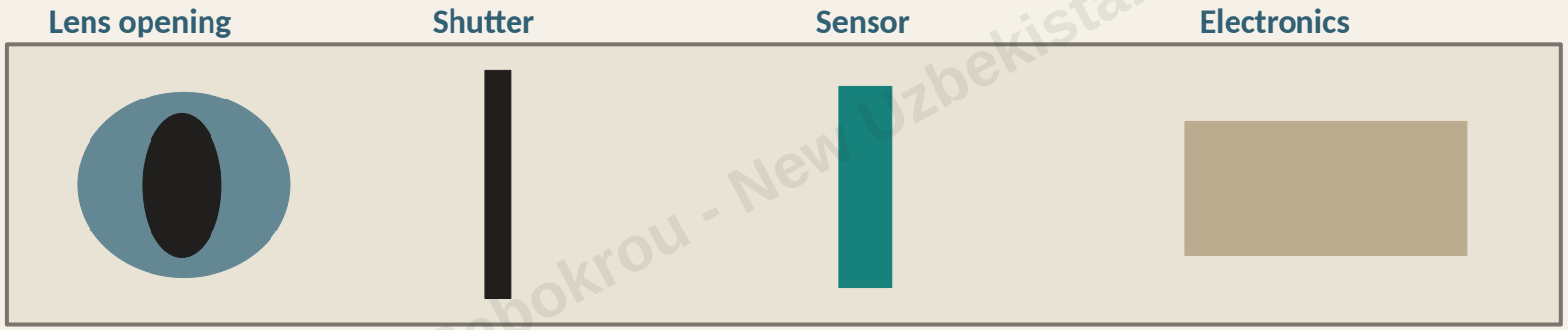
Underexposed | Balanced | Overexposed

Example: with the same light and lens opening, collecting light for twice as long doubles the exposure.

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Inside the camera: light, time and gain

Aperture and shutter change the light collected. Gain changes the electrical signal.



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Aperture

Wider opening:
more light.
Background may blur.

Shutter time

Open longer:
more light.
Motion may blur.

Sensor

Collects light
as electrical charge.

Gain / ISO

Amplifies the signal.
Adds no photons.
Noise may be clearer.

Main idea: farther 3D points appear closer to the image centre

$$u = \frac{fX}{Z} \qquad v = \frac{fY}{Z}$$

X, Y and Z locate a point in the real world relative to the camera. f describes the lens. The equations calculate u and v, the point's 2D position on the image plane.

WORLD → IMAGE

X = left/right, Y = up/down and Z = distance forward in the real world. f is focal length. u and v are the resulting 2D image coordinates.

MAIN RESULT

Because Z is in the denominator, increasing distance Z makes u and v smaller: the point moves toward the image centre and the object appears smaller.

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PREDICT

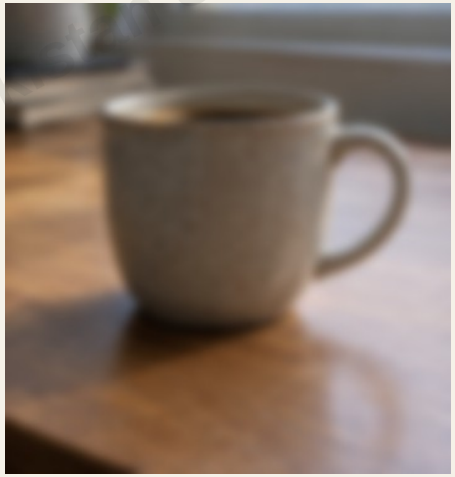
Should an object label change when the object moves farther from the camera?

Why do edges look blurry?

A sharp edge changes suddenly. Blur spreads that change over neighbouring pixels.



Sharp: the cup rim is easy to locate.



Blurred: the rim spreads into a soft band.

Incorrect focus or camera motion can cause blur. Blurring does not move the actual cup.

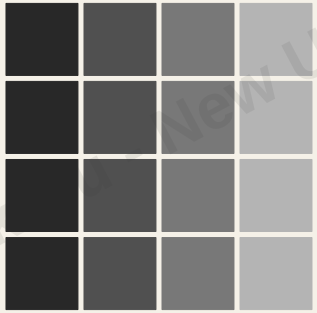
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Sampling: choosing the image grid

Main point: a digital image measures the lens image at fixed grid positions. Each position becomes one pixel.



1. Lens forms a continuous image



2. Sensor chooses 16 positions

3. Store one number per position

I[row, col] = pixel value
row: vertical position
col: horizontal position
I[2,3] = 120

More positions = more spatial detail.
Too few positions can miss thin lines or create false-looking patterns.

Sampling chooses locations. Quantisation later rounds each measured value to an allowed number.

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Pixel intensity and perceived brightness are related, not identical

In an image-formation model, a code is a sensor response. Brightness is an interpretation that also depends on display, surroundings, and the observer.

STORED MEASUREMENT

The centre patch can have exactly the same RGB code in two images.

[153, 153, 153]

PERCEIVED BRIGHTNESS

Surrounding context can make identical measurements look lighter or darker.

Human inference uses context

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Same measurement, different perception. The array has no surround context unless an algorithm represents it.

Why grayscale uses different RGB weights

Grayscale combines RGB into one brightness value. Green contributes most because human vision is more sensitive to it.

$$Y = 0.2126R + 0.7152G + 0.0722B$$



Original colour



Grayscale: colour differences disappear.

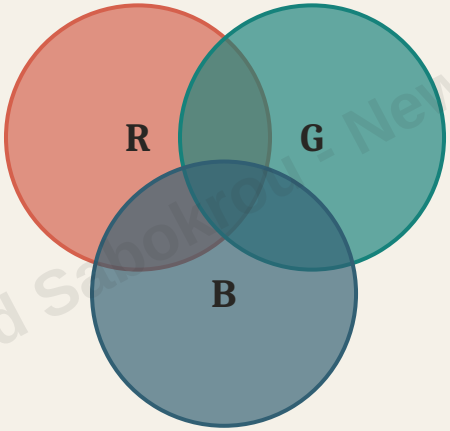
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For linear RGB (0.2, 0.6, 0.1): Y = 0.04252 + 0.42912 + 0.00722 = 0.47886.

These standard sRGB weights sum to 1. Undo sRGB encoding before applying them; encode Y again to display it.

RGB stores three sensor responses at each pixel

Displays add red, green, and blue light; cameras measure light through three broad filters. The values are sensor responses, not universal colour names.



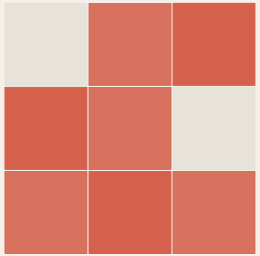
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$$I[r, c] = (R, G, B)$$

One coordinate holds one red, green, and blue measurement; 8-bit values commonly lie in [0, 255].

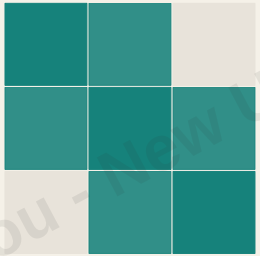
One colour image is three aligned matrices

RED CHANNEL



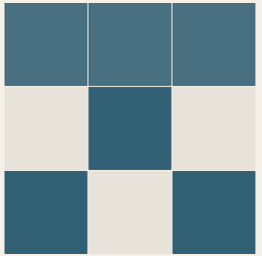
I[:,:, 0]

GREEN CHANNEL



I[:,:, 1]

BLUE CHANNEL



I[:,:, 2]

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Alignment is essential: the three values at one coordinate describe one sensor location. Misalignment creates colour fringes and corrupts colour differences.

RGB values depend on camera, illumination, and the processing pipeline

RGB is an encoding of three broad sensor measurements. It is practical for storage and display, but its numbers do not have a device-independent “true colour” meaning.

CAMERA SENSITIVITIES

Different cameras use different red, green, and blue filters, so the same spectrum can yield different triples.

WHITE BALANCE

The camera may multiply channels to make a neutral scene look neutral under an assumed illuminant.

TONE MAPPING + JPEG

A saved image can contain tone mapping, gamma, compression, and clipping rather than raw linear sensor values.

CONSEQUENCE

Do not call an RGB triple the object’s “true colour” unless the camera, illumination, and processing conditions are specified.

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SELF-STUDY CHECK

When comparing images, record the capture pipeline if possible. A colour threshold learned on one camera or lighting setup can be fragile on another.

Different coordinates make different visual questions easier

RGB

STORAGE + DISPLAY

Three device channels

HSV

INTERACTIVE THRESHOLDS

Hue, saturation, value

Lab

COLOUR DIFFERENCE

Lightness + opponent axes

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A colour-space conversion changes which relationships are simple to express. It rearranges information; it does not create it.

HSV describes colour in three ways

H = colour family. S = colour strength. V = the largest RGB component, a brightness-like value.



Original photo



H: colour family



S: colour strength



V: maximum RGB

Example pixel	H (degrees)	S	V
Brown/orange	30°	0.75	0.80

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Converting RGB to HSV: a small example

RGB and HSV describe the same pixel with different coordinates. HSV can make colour selection easier.

R	G	B
0.8	0.5	0.2

H	S	V
30°	0.75	0.80

M = max(R,G,B) = 0.8
m = min(R,G,B) = 0.2
Δ = M − m = 0.6
V = M = 0.8
S = Δ / M = 0.75

Red is the maximum in this example:
H = 60° × (G − B) / Δ = 30°

Benefit: select orange by hue, then reject nearly grey pixels using S.
Hue is unreliable when S is near zero.

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Use cv2.cvtColor(rgb, cv2.COLOR_RGB2HSV). For uint8 OpenCV: H = 0–179, S and V = 0–255.

White balance: making a white object look neutral

Example first: a white cup looks yellow when the blue channel is too weak.



Before: simulated warm colour cast



After: channel gains restore the reference.

$$I'(c) = g(c)\, I(c)$$

c = R, G or B; I = input; g = channel multiplier; I′ = output.
White patch: (200,160,120) × (1,1.25,1.67) ≈ (200,200,200).

Choose gains using a known neutral reference. This adjusts colour balance, not the material itself.

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The same object under different lighting

The cup stays the same. Its recorded colour changes when the lighting changes.



Warm light: yellow tint



Neutral reference



Cool light: blue tint

Colour constancy means recognising the same surface colour despite lighting changes. White balance helps, but mixed lighting can defeat one correction.

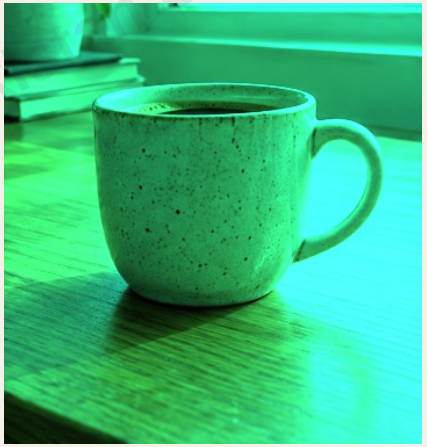
When can RGB channels change differently?

Both examples can happen without changing the object. The numbers alone cannot prove the cause.



A: increase all channels by about 1.46.

A: (110,82,61) becomes (160,119,89).
Example: brighter illumination or higher gain.



B: strengthen green and reduce red.

B: (110,82,61) becomes (41,156,78).
Example: green lighting or wrong white balance.

Shown as simulated channel gains. An object change is another possible explanation.

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5-MINUTE BREAK

Light created the numbers. Now read the distribution.

When we return: histograms, contrast, clipping, and the limits of distribution-only features.

Keep the opening question in mind: where did the grey card become blue?

A histogram counts how many pixels fall into each intensity bin

$$h(b) = |\{p : q(I(p)) = b\}|$$

Count pixels p whose quantised intensity q(I(p)) belongs to bin b. The count does not retain p's location.

WHAT IT KEEPS

Brightness distribution, clipping, broad contrast, and—when extended—some colour distribution statistics.

WHAT IT LOSES

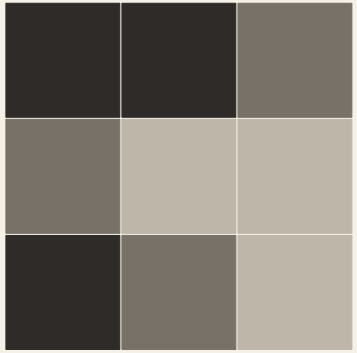
Pixel location, object shape, boundaries, and the causal source of a value in the image.

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SELF-STUDY CHECK

Reading check: a permutation of pixel locations leaves the histogram unchanged. It is a distribution feature, not a spatial representation.

Build a histogram by counting a small image





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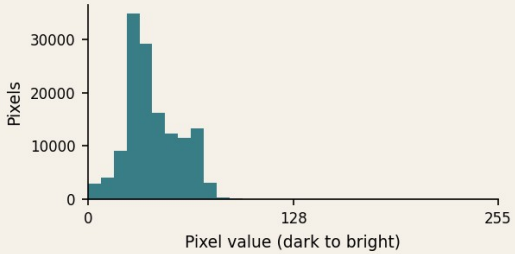
4 bins: 0, 1, 2, 3

The matrix and histogram contain the same number of pixels. The count survives; the spatial arrangement does not.

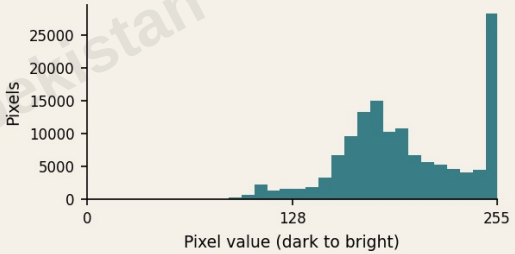
What do these histograms suggest?

For each graph: mostly dark, mostly bright, low contrast, or a broad range? Explain your choice.

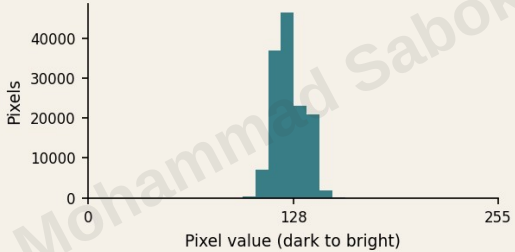
A



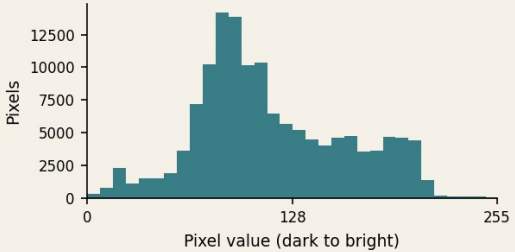
B



C



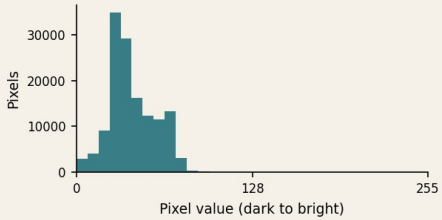
D



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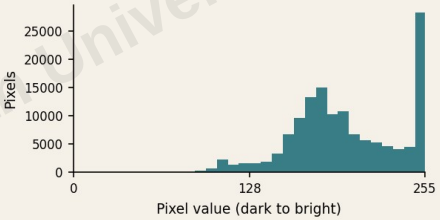
Histogram answers: dark, narrow or broad?





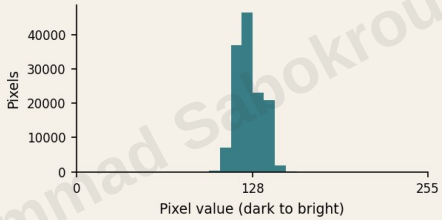
A: dark; values cluster left.





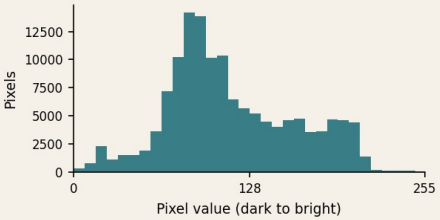
B: bright; a spike at 255 means clipping.





C: low contrast; values are close together.





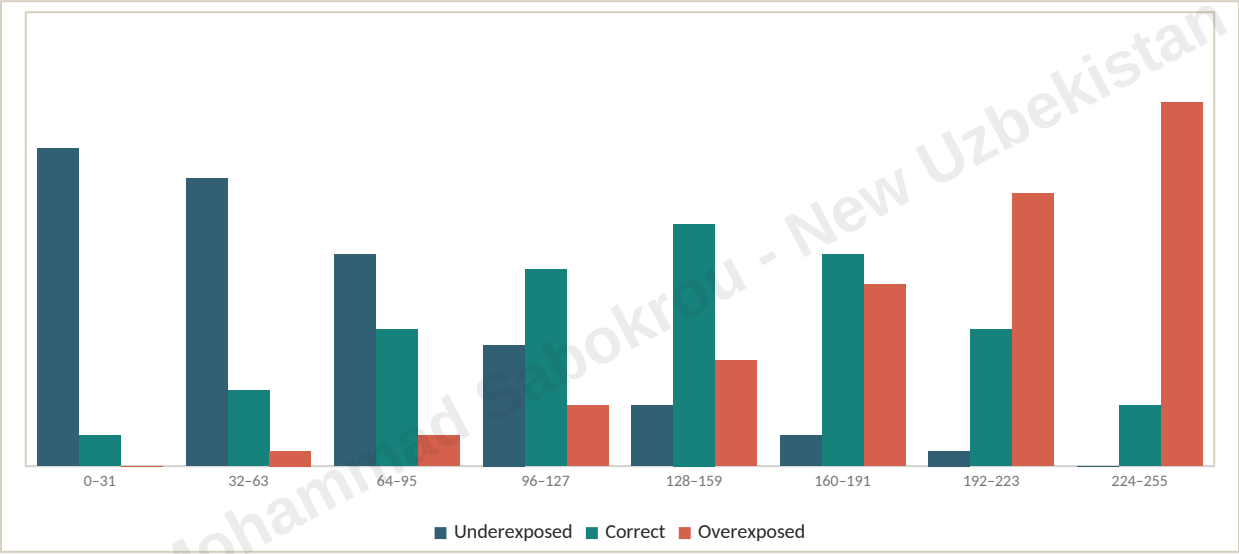
D: broad range; dark and bright values.

A broad histogram does not guarantee a useful photo: a sharp face and its shuffled pixels have the same histogram, but only one is recognisable.

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The distribution moves when exposure changes

Illustrative intensity-bin counts for the three exposure conditions



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READ THE TAILS

The leftmost bin warns about crushed shadows. The rightmost bin warns about clipped highlights.

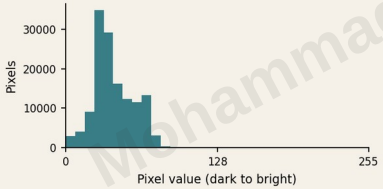
This is a measurement aid, not proof that one exposure is universally best.

Which change loses image detail?

Judge the loss of useful detail, not brightness alone. Histograms below come from the displayed images.

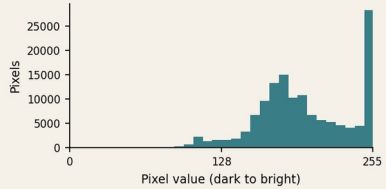


A: dark, but cup shape remains



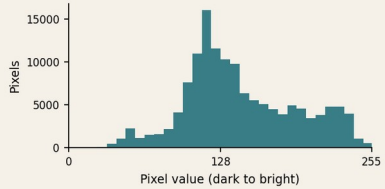


B: clipped highlights lose texture





C: add a constant to every pixel



B is the clearest warning: many bright values merge at 255. A may retain detail; C also loses detail wherever the added offset clips.

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A transformation changes pixel values

A transformation is a rule that replaces each old pixel value I with a new value J.



Input I



Output J = clip(1.4 I − 20, 0, 255)

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J = αI + β: α multiplies differences (contrast); β shifts all values (brightness).
For I = 100, α = 1.4 and β = −20: J = 120. Clip results to 0–255.

Gamma changes dark and bright values differently

Use gamma to reveal shadow detail or darken midtones. Divide 8-bit values by 255 first.



γ = 0.5: lifts dark values



γ = 1: unchanged



γ = 2: darkens middle values

$$J = I^{\gamma}, \qquad I \in [0, 1]$$

I = input; J = output; γ (gamma) = chosen exponent.
At I = 0.25: γ = 0.5 gives 0.50; γ = 2 gives 0.0625.
Multiply J by 255 to display on the 8-bit scale.

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Contrast stretching makes a narrow range wider

Contrast is the difference between light and dark values. A narrow range can make a photo look flat.



Before: values roughly 102–153



After: expand that range to 0–255.

J = 255 × (I − low) / (high − low), then clip to 0–255.
I = old value; J = new value; low = 102; high = 153. At I = 128, J ≈ 130.

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CDF: a running fraction of pixels

CDF means cumulative distribution function: the fraction of pixels with value less than or equal to a chosen level.

Level l	Count	Running count	CDF
0	1	1	1/10 = 0.10
1	3	1+3 = 4	4/10 = 0.40
2	4	4+4 = 8	8/10 = 0.80
3	2	8+2 = 10	10/10 = 1.00

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There are 10 pixels. CDF(1) = 0.40 means 40% of them have value 0 or 1. We can use this fraction to choose a new brightness value.

Equalisation uses the CDF to change brightness values

For each old brightness I, the CDF tells what fraction of pixels are at or below I. That fraction chooses the new brightness J.



Input: narrow brightness range



After equalisation: differences are clearer.

1. Choose I 2. Read CDF(I) 3. Multiply by L − 1 4. Round to get J

$$J = (L - 1)\ \mathrm{CDF}(I)$$

I = old brightness; CDF(I) = cumulative fraction; L = number of allowed values; J = new brightness.
Example: L = 256 and CDF(I) = 0.40 gives J = round(255 × 0.40) = 102.

Pixels with the same old value receive the same new value. Equalisation cannot recover clipped detail.

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More contrast can reveal detail and noise

Contrast means differences between dark and light. Making these differences larger can reveal both detail and unwanted speckles.



Before: flat image with small noise



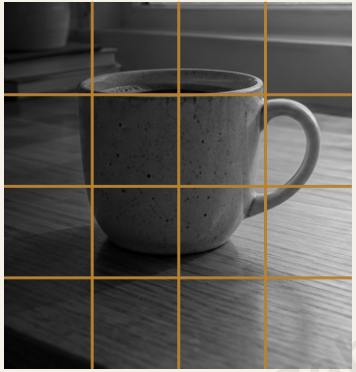
After: cup texture AND noise stand out.

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Benefit: cup details are easier to see. Cost: speckles become more visible too. More contrast cannot tell useful structure from noise.

CLAHE improves contrast in small image regions

CLAHE = Contrast Limited Adaptive Histogram Equalisation. Use it when different parts of an image need different contrast adjustments.



Input divided into 4 × 4 regions



Local contrast improved; detail is easier to see.

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Tiles

Small image regions

Local histogram

Counts in each region

Clip limit

Limits tall histogram peaks

Blend

Joins neighbouring results

The grid marks image regions. The four boxes describe processing steps. Strong settings can still amplify noise.

Why a colour histogram needs many bins

A bin is a group of similar values. A 3D colour histogram counts combinations of red, green and blue groups.

16 bins for red × 16 for green × 16 for blue
= 4096 possible colour combinations

With 8-bit RGB, each bin covers 16 values.
Example RGB (105,60,30) falls in bins (6,3,1), counting from 0.
We add 1 to that combination's count.

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Compared with 16 grayscale bins: more colour detail, but 256 times as many counters. With only 100 pixels, at least 3996 bins must stay empty.

Two images can share a histogram and differ in spatial structure

Image A

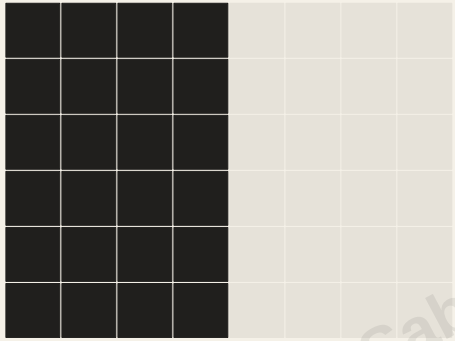
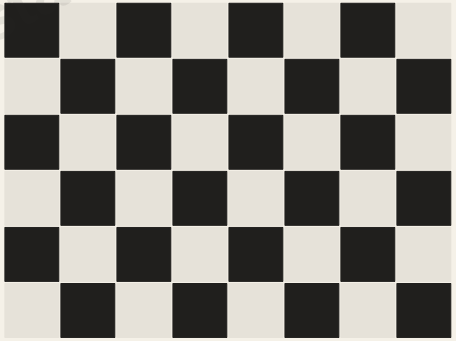




Image B



Each has 24 dark and 24 light pixels.

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A histogram counts values but cannot identify an edge, boundary, or object location.

Raw pixels and histograms keep different evidence

A representation choice decides which changes can be ignored and which distinctions remain available for the task. It is an evidence decision, not just a data format.

RAW PIXELS

Keep location, geometry, and local texture. Sensitive to shift, light, and alignment.

Strong for: localisation

HISTOGRAM

Keep a distribution of values. Insensitive to pixel permutation, but discard layout.

Strong for: exposure diagnosis

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Choose the representation from the invariance you need and the distinctions you must preserve; validate the choice with a task metric.

Measure clean and altered conditions separately

A robustness claim needs paired evidence: hold the task and metric fixed, change one recording condition at a time, and report the shift explicitly.

CLEAN REFERENCE IMAGES

Report accuracy or error under the intended baseline condition; retain the same split and metric for every comparison.

DIMMED IMAGES

Report the same metric after a controlled exposure shift, then inspect whether shadow detail or noise explains the change.

COLOUR-CAST IMAGES

Report the same metric after a channel-specific shift; compare the raw input and any white-balance repair.

CLIPPED IMAGES

Report the same metric after highlights or shadows lost information. Do not describe clipping as a recoverable colour cast.

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SELF-STUDY CHECK

Compare the original, the changed image and the processed result. State which details improved and which were lost.

What to remember

The image is a measurement. Understanding how it was made helps us interpret its numbers.

Light, surfaces and camera settings affect pixel values.

RGB, grayscale and HSV describe different aspects of the same image.

Histograms count values but do not preserve object shape or location.

Image transformations can reveal detail, but can also clip values or amplify noise.

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